📝 Notes

Mar 2, 2026

RH & Pansonic Avionics sync

Invited Bill Venella Phillip Eurs Bill Stoker Jason Anderson Sunil Gaddam Jason Dobyns

Attachments RH & Pansonic Avionics sync Notes - RH & Pansonic Avionics sync

Meeting records Transcript

Summary

The meeting participants, including Bill S. (he/him), Jason Anderson, Bill Venella, and Jason Dobyns, primarily discussed the features and implementation of Red Hat Enterprise Linux 10 (Rel 10), which focuses on AI integration, enhanced security with quantum cryptography, and the introduction of "image mode" for treating the OS as an immutable container to streamline deployment and rollback capabilities. Jason Anderson detailed how Rel 10 addresses skill gaps and security supply chain needs, confirming that AI-assisted features like "command line C" support both connected and air-gapped environments, and committed to future discussions about containerization, legacy system migration, and Kubernetes, for which Bill Venella suggested a dedicated follow-up meeting after gathering necessary information. Jason Dobyns intends to begin implementing Rel 10 by integrating it into their existing Packer project for testing installation methods and package naming conventions.

Details

- **Meeting Logistics and Participant Check-in**: The meeting started with audio checks and participant greetings, noting that Philip would not be joining due to illness (00:00:00). The team decided to focus the discussion on Red Hat 10 (Red Hat Enterprise Linux 10, referred to as "Rel 10" in the transcript), delaying a deeper dive into Kubernetes topics for a subsequent meeting (00:03:06). Bill Venella suggested collecting the required Kubernetes information to bring on experts for a dedicated call (00:03:55).
- **Participant Status and Note Sharing**: Bill S. (he/him) informed the group that they would be multitasking due to a need to arrange transit for US team members currently stranded in Dubai. Jason Anderson assured Bill S. (he/him) that they would send them the notes and the slide deck afterwards (00:04:57).

- **Overview of Red Hat's Evolution**: Jason Anderson provided a brief history of Red Hat, noting their presence in the market for over a quarter-century, starting from their initial release in 1994, focusing on Unix to Linux, and evolving through virtualization and containers, with the next evolution being AI. They also highlighted Red Hat's extensive reach across secure companies, telecom, commercial banks, and their large partner ecosystem (00:05:38).
- **Red Hat 10 Focus Areas and Market Needs**: Red Hat 10 is concentrating on key areas, including enabling organizations to do more with less, preparing for cloud adoption, enhancing security and speed, and integrating AI capabilities. They noted that Rel 10 is designed to address skill gaps, making tasks easier for administrators through features like command line assist (00:06:57).
- **Introduction to Image Mode**: The concept of "image mode" was introduced, which treats the operating system as a container, allowing users to build images for bare metal, cloud, or virtualization environments using a CI/CD pipeline. Jason Anderson clarified that this feature is not intended to replace existing tools like Packer (00:08:17).
- **AI-Assisted Decision Making and Security Enhancements**: Red Hat 10 introduces AI to aid in decision-making during processes like image building, suggesting related packages based on common customer installations, which helps save time and streamline the process (00:09:16). Additionally, Red Hat is the first to introduce quantum cryptography, which is FIPS compliant and designed to withstand quantum computing, providing a crucial security layer that can be enabled with a simple flag at install time (00:10:38) (00:36:58).
- **Support for Data Science and Efficient Deployment**: Rel 10 provides a platform for data scientists to run import analysis and work with AI models as a proof of concept on standard x86 hardware without immediately requiring a massive investment in a huge platform (00:10:38). The system is designed to streamline computations and allows users to build and deploy images efficiently across various platforms such as OpenShift, VMware, AWS, and Azure (00:11:55).
- **New Command Line Assistance Feature**: A new application called "command line C" (likely "command line Assist") is included to help users by answering basic questions about Linux, issuing commands, or performing routine tasks (00:11:55). The utility can even generate a bash script for complex commands, such as the `find` command, which users should review before running (00:13:23).
- **Command Line Assist Deployment Model**: Bill S. (he/him) inquired whether the AI-assisted features require internet connectivity or utilize a local knowledge source (00:13:23). Jason Anderson confirmed that the feature supports both connected and air-gapped environments, recommending the use of Satellite to import the necessary data into a local model (00:14:26).
- **Red Hat Light Speed Vision and Automation Tools**: Red Hat's vision, referred to as "Linux light speeded vision," focuses on streamlining processes, making users

faster, more proactive, and better at their jobs. The system provides tools that support complete end-to-end CI/CD automation, integrating with existing tools and helping to troubleshoot issues like why SSHD is not starting by checking versions and retrieving logs (00:14:26).

- **Image Mode for Simplified Operations and Security**: Image mode is designed to work like a smartphone where updates that go wrong can be immediately rolled back to the previous version, removing the fear of upgrades and continuous testing (00:16:50). The operating system is treated as an immutable container, which prevents unintended changes to the system, thereby strengthening security (00:19:12).
- **Cybersecurity Collaboration and Automation Principles**: Jason Dobyns mentioned that their organization has a cyber security team rolling out tools like CrowdStrike, and they expressed interest in facilitating a conversation between Jason Anderson and their security team (00:20:20). Jason Dobyns and Jason Anderson agreed on the principle of automating any task that must be done more than once (00:21:18).
- **Building Images with Image Mode and Container Management**: Image mode allows users to create consistent, non-snowflake environments that enforce security and compliance policies (00:23:23). The configuration uses a file structure similar to Docker or Podman build files, specifying the base image, packages, application layers, and configuration files. For managing individual containers, Jason Anderson suggested using traditional Rel 10 with a utility like Podman Desktop, or discussing Kubernetes for management at scale (00:24:25).
- **Kubernetes and Service High Availability**: Jason Dobyns discussed their positive experience with Kubernetes' ephemeral nodes, which simplifies updates by allowing replacement of hosts with minimal downtime (00:25:51). They detailed how implementing health checks and running multiple replicas across multiple physical hosts mitigates service interruptions during updates and ensures production readiness (00:26:57).
- **Kubernetes Upgrade Risks and Mitigation**: Jason Anderson cautioned that increasing replica counts during a Kubernetes upgrade could potentially break the process if the scheduler places replicas on the same host and resources become utilized during the draining of a node. Jason Dobyns countered that implementing node selector labels can help ensure replicas are spread across different nodes (00:28:56).
- **Red Hat Insights and System Management**: Jason Anderson inquired if the team uses Red Hat Insights, which is being rebranded as "Lightseed" (00:30:03). This free tool provides a "single pane of glass" for administering all registered systems via the hybrid console, allowing them to view packages, system information, vulnerabilities, and push updates (00:32:13).
- **Security Supply Chain and Image Compliance**: Jason Anderson stressed that Rel 10 helps organizations meet security and compliance standards by providing

better decisions at build time and hardening images, which makes it easier to document and show the security supply chain to compliance officers (00:33:24). This streamlining helps rectify images and roll them out faster when dealing with vulnerabilities (00:34:34).

- **Next Steps for Rel 10 Implementation**: Jason Dobyns plans to start implementing Rel 10 by incorporating it into their existing Packer project to test installation methods and package naming conventions (00:39:20). They also asked about the upgrade path from Rel 10 to future versions, which Jason Anderson confirmed should be easy (00:40:23).
- **Discussion of Legacy Systems and Future Planning**: The group discussed challenges with legacy CentOS 7 and other older Linux machines running applications that require older Java versions, necessitating long-term support (00:40:23). These applications will be targeted for containerization and migration in the next corporate year, which starts in April (00:42:24). Jason Anderson expressed interest in being involved in those discussions and offered assistance (00:43:26).
- **Resource Sharing and Kubernetes Follow-up**: Jason Anderson shared a link to the Rel 10 UBI (Universal Base Image) as a starting point for building a golden image (00:44:38). Bill Venella needed to leave the call and suggested setting up a follow-up meeting to discuss Kubernetes interests. Jason Dobyns outlined their plan to finalize the CrowdStrike deployment to AKS (Azure Kubernetes Service) and complete a production readiness review before having a more beneficial discussion about OpenShift and their current AKS setup (00:45:36) (00:47:46).
- **Commitment to Agenda Setting**: Jason Anderson requested that the team provide an agenda for future meetings to avoid unprepared discussions, committing to bugging them a week out for topics (00:46:50). They reiterated their goal to make the team's life easier and ensure they find value in the meetings (00:47:46).

Suggested next steps

Jason Anderson will send an email or bug the group a week out before the next meeting to discuss potential topics for the next call.

Jason Anderson will send notes and the slide deck to Bill S. (he/him) after the meeting.

Jason Dobyns will provide Jason Anderson with contact information or an introduction to the cyber security team for a conversation about security.

Jason Dobyns will try out Red Hat 10, using the provided UBI image as a starting place for building a golden path image with the packer project and checking out Red Hat Insights (Lightseed).

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00:00:00

Jason Anderson: Uh, if you're speaking, I can't hear you. How's everybody doing?
Bill S. (he/him): I don't know if you can hear me. I'm trying to figure out what's going on with my audio and video.
Jason Anderson: I can hear
Bill S. (he/him): Okay. Um, I've got to figure out why it's not seeing my other
Jason Anderson: you.
Bill S. (he/him): camera.
Bill Venella: Hey guys, good
Sunil G: Hey
Bill Venella: morning.
Sunil G: is.
Jason Anderson: That's just about everybody. Is Philip gonna join? Do you guys know?
Sunil G: Yep.
Jason Anderson: No.
Sunil G: No.
Jason Anderson: Okay. Uh, well, how's everybody doing? It's been it's been a a week or so since we
Jason Dobyns: Good morning.
Jason Anderson: talked.
Jason Dobyns: Yeah,
Sunil G: Thank Yep.
Jason Dobyns: Phil's not going to join us because he's not feeling well, but uh so yeah,
Jason Anderson: Okay.
Jason Dobyns: and uh I I forget what we were we're going to go over uh Red Hat 10 or RA
Jason Anderson: Yeah. Yeah.

00:03:06

Jason Dobyns: 10.
Jason Anderson: Roen was the the thing I threw out there unless somebody else wanted to discuss something else, you know,
Jason Dobyns: Um I I don't know about that. I mean,
Jason Anderson: different.
Jason Dobyns: Real 10 is something we'll be we'll appreciate to see. And then I know our talk last time kind of went to some of the Kubernetes things that you guys have to offer as well. So,
Jason Anderson: Yes.
Jason Dobyns: um we'll absolutely let's talk about real 10 first and if we still have time then we can go down that path as
Jason Anderson: If you want to talk about Kubernetes,
Jason Dobyns: well.
Jason Anderson: I'm more than happy to switch and talk about Kubernetes. Um, it's a you this is your meeting. I want you to get the most value out of
Bill Venella: Well,
Jason Dobyns: I think I think Rail 10's probably the easy or not the easy the quickest one for us to be able to jump into because
Bill Venella: so
Jason Anderson: it.
Jason Dobyns: like I said we have a project that runs packer jobs.

00:03:55

Jason Dobyns: Um and I'm doing I'm doing uh a few different ver versions of Linux in there. And so adding going up from rail 9 to 10 with that project will be pretty easy on our end. Uh the Kubernetes stuff is definitely a longer set of work that will most likely have to
Bill Venella: Yeah, my my suggestion is let's collect uh what you're looking
Jason Dobyns: happen.
Bill Venella: for from a Kubernetes and we'll bring on some experts on the next topic call. Uh then we can go through all of that.
Jason Dobyns: Sure.
Bill Venella: We can we can provide more detail.
Jason Anderson: Absolutely. All right, then. Uh, you guys want to get into real 10 or is there any other questions you would like to ask or before we get into this?

Jason Dobyns: I don't think I have questions just yet or much at all. So,
Bill S. (he/him): And I just I just want to be really transparent.
Jason Anderson: All right.
Bill S. (he/him): I'm going to be multitasking a little bit. I'm not too in the weeds with many of the things the Linux team is doing.

00:04:57

Bill S. (he/him): And I'm uh we've got some people stranded in uh Dubai right now from our US team that we're trying to arrange transit for. So I'm going to be looking like I'm not paying attention. You'll I will be listening. Um and it's not because I'm not interested in the content. It's that I'm trying to make sure my people can get safely out of the Middle East while bombs are
Jason Anderson: Yeah,
Bill Venella: No,
Bill S. (he/him): flying.
Jason Anderson: that is definitely the priority and I will be happy to send you notes and the slide
Bill Venella: it works.
Jason Anderson: deck afterwards.
Bill S. (he/him): I'll get debriefed by my team.
Jason Anderson: Uh
Bill S. (he/him): I'll they'll they'll they'll my team tells me what they need me to do.
Jason Anderson: okay,
Bill S. (he/him): That's I have a great team, so it's very easy to manage them. I just do what they ask me to do and get out of their way.
Jason Anderson: sounds good. All right. Well, here we go.

00:05:38

Jason Anderson: Red Hat 10. Um, so, uh, just wanted to start out, you know, Red Hat's been doing this for a while. Um, over quarter century to be in fact. Um, our initial release came out in 1994. Uh, you know, we were really trying to do the Unix to Linux thing. um you know that ran up a pretty good way until about 2002 and then we started switching more to um you know virtualization side of things. Um we you know started our longer project life cycles all of those fun things. Um, but then as it's evolved, we're now moving into containers and then the next evolution is AI, right? I mean, I think everybody would agree that that's where we're headed at this point. Um, so tried, true, tested, we're everywhere. Fortune 500, all your secure um companies, telecom, commercial banks, you name it, we're there. Um, we have a huge partner ecosystem. Um, you know, Red Hat is been everywhere. Try, true, and tested. All right, here's where it starts to get interesting.

00:06:57

Jason Anderson: Um, what are we trying to do with Red Hat 10? Um, you can see the four process or thoughts we have here. Um, everybody is trying to do more with less. Everybody is trying to prepare for you know going to the cloud as well as trying to do it all as being as secure as possibly as fast as possible and then oh yeah let's throw in AI on it so we can you know just make it a little bit easier and you know have these big huge complex models that take up lots of resources and are you know very interesting for data scientists. Um but it's you know it's where we're what Real 10 is focusing on. Um so some of the things that you know I mentioned we'll be addressing skill gaps. Uh you know we want everybody to be as fast as they can doing whatever task that they're trying to do. Um well, you know, for the hiring of administrators, we we make things easier like with command line assist and to be able to get help and information about Linux and you know, help helping with overall system administration.

00:08:17

Jason Anderson: Um we also introduce a uh new concept called image mode. I don't know if you guys have heard of it or played with it at all. I believe you told me you had played with it a little bit last time we spoke, J. And
Jason Dobyns: Yeah, like it got demoed for us sometime last year,
Jason Anderson: uh
Jason Dobyns: I think, just a little bit and and it was part of a couple of the conversation calls we had.
Jason Anderson: okay. Um well, you know, I'm just going to highlight on it a little bit because I'm a huge fan. Um but as you know, it is we're now treating the operating system as a container. Um, so you can build your images whether it be for a traditional brow metal system uh machine in the cloud. I don't care what cloud uh your virtualization environment you can build all those packages. I'm not trying to replace

packer. I'm just telling you what what image builder can do for you. Um all in a CI/CD pipeline.

00:09:16

Jason Anderson: So you are really taking that step left to a CI/CD model. Um, we also introduce AI to help you at decision time. And what this is is um when you're going through like our image builder uh tool and you say I want to install Engine X, right? It's going to come back and say, I've seen all of these other customers when they install EngineX, they install all these other packages, too. Would you like to install those packages? Just little things like that to make it easier, make sure you don't forget. Um, but you know, save time, streamline the entire process, but do it better the first time. Um, security. This one's very interesting. Um, I don't know how much data you guys store, um, or where you store it or anything like that, but what I do know is that 90% of the data that has been hacked, pulled from some other system, whether they know it or not, is sitting out there on the internet waiting for machines that have the inc power to break the encryption that is being used.

00:10:38

Jason Anderson: Right? This is just a known fact. The data is out there. They're just trying to figure out how to get to it. Uh Red Hat's the first one to introduce quantum cryptography. Uh FIPS compliant um very forward thinking and is designed to withstand quantum computing. Um, this is kind of a conundrum to me with IBM being one of the leading uh drivers for quantum computing that were designing the cryptography for it. But it it is uh where we're at and like I said, FIPS compliant and I would highly highly recommend doing it just to save yourself a little bit of headache. It's a simple flag um at install time. Okay, sorry, Bill. Um, AI, you know, we've kind of t talked about that if you have any data scientist on board, any GPUs that you're doing any type of import analysis, uh, this is the great way for them to get started without having to invest in a huge platform to run your AI on and all of those things. So, think of this as your proof of concept ground.

00:11:55

Jason Anderson: Um we have done extensive work with this to streamline the uh process for computations in doing these models but once again you are working on standard x86 hardware of no GPU so don't expect a miracle but it is streamlined and much more efficient. Um you and I also mentioned you know I can build your image or you can build your image for whatever workload you're liking. Uh whether it be you know a preconfigured container you know in the cloud um you know Open Shift Nanics VMware AWS Azure you name it. Uh, we have a way to deploy Relin out there efficiently, quickly without caring where it's running. Oops, excuse me. Got off track there. All right, so kind of mentioned uh addressing the skill graphs. Uh, we have a new application called command line C. Uh this allows you to ask basic questions uh about Linux or how to issue a command or um do routine task even. Uh the the great example I always use I might have mentioned it last time is the find command.

00:13:23

Jason Anderson: I don't know too many people they can remember that syntax off the top of their head on the fly. This is a great resource for things like that of how would I go about doing that? It will even write you a bash script to do that that you can write have for later. Um, obviously I would highly suggest reviewing it before you're running it. I don't trust AI that much, but these are the advances that we're doing. Once again, streamlining the process, make it faster, all of those things that I mentioned at the beginning. Um, Red Hat L.
Bill S. (he/him): Quick question on that last one. Uh on the AI,
Jason Anderson: Yes.
Bill S. (he/him): is that require internet connectivity or is that using a local knowledge source on the
Jason Anderson: either
Bill S. (he/him): VM?
Jason Anderson: or. So you can um bring it into an airgapped environment. Uh the recommended method would be to use satellite um and then import that into a model or something like that. Um but there are ways to do it in and out.

00:14:26

Jason Anderson: You can use your own models. We don't really care. It's just the data you got to bring it into your model.
Bill S. (he/him): Thank you.
Jason Anderson: Yep. All right. Uh we talked about light speeded. Uh ah the Linux light speeded vision. Uh it's all about you know what I said earlier of streamlining it, making it better. We want to make you faster. We want to make you proactive. and we want to make you better at your job. Um, that is our goal. That is our whole purpose behind real 10, you know, I believe. And you know, we have lots of different tools um that help and can tie into the Red Hat product um that you can do complete end toend CI/CD automation. And I know you've mentioned Terraform um but you know with all the Red Hat tools you can lock all of that together in one platform. Uh I mentioned the command line assistant and looking for uh helping you troubleshoot.

00:15:36

Jason Anderson: So as you can see here it's uh why is SSHD not starting? Um it can go through and check the version, get your logs for you. Um, it can even go as far as to uh tell you what is wrong and what you should fix. Um, once again, streamline. Let's use the AI. Let's use the knowledge. Let's just make things faster. Excuse me. Um, once again, I keep talking about DevOps, CI/CD, every, you know, shifting left. I believe that is the right mindset. Let's keep developing faster, deploying faster, getting our product out there and making sure we're getting the best thing out there that we possibly can, right? That that's everybody's end goal. So, let's just make that easy. Streamline the oper, you know, how we do it with the tools, you know, that we've talked about for building your images, deploying your images. We're also increasing your security just with, you know, the encryption out of the box. Plus, you've got the decades of Red Hat security built on top of this, you know, and once again, we're just making it all easier.

00:16:50

Jason Anderson: That that's the goal is to get to that place easier and faster, saving you both money, time, and efficiency on your hardware. This is to me uh the coolest slide that describes uh image mode is I want it to work like my smartphone. What? What do I mean? I mean an update went wrong. What do I do? I reset. Roll back. I'm going to turn in my phone. I don't want anybody to have my data on there. What do I do? I reset my phone. It goes back to the original day that I bought it. No information, no nothing. This is image mode, right? Yes, you can deploy an image. And yes, you can even apply updates to it. You can do things like, you know, update your Apache, your engine X, you know, whatever services that you would issue your normal pseudo command to restart set service. Um, what you can't do is all your complete kernel upgrades.

00:17:52

Jason Anderson: That will require a new image, a new roll out. But, you know, like I said, that is where we are. And if that goes wrong, we immediately roll back to the previous version. So once again, now that fear of doing those upgrades and testing o all those upgrades continuously and improving the overall performance all goes away. If I screw up, I just roll back to the original version. Life is good. It kind, you know, it it's a a very nice, how do I put this? Uh security blanket is the way I would describe it. Does anybody have any questions about any of this stuff? I mean, is any of it resonating? um with you
Jason Dobyns: No,
Jason Anderson: guys.
Jason Dobyns: I mean it image mode sounds like kind of like a container type path forward
Jason Anderson: Absent.
Jason Dobyns: for something outside of like a Kubernetes type deployment where you're everything's a container.
Jason Anderson: Yes.
Jason Dobyns: So it makes it part of core parts of the OS are now containerized which the main part the main thing that I like about Kubernetes in general is uh all the systems behind it are ephemeral in my like they they they can all be easily replaced at any time and so having images for parts of the core part of the OS kind of tells me that's sort of the path they're they're trying to head towards.

00:19:12

Jason Anderson: Yep. Yep. And now you can layer on your application on top of that as you go through that entire process with automation. Um, you're spot on with that. But I'm going to point out another side that is just as important and often overlooked. You mentioned that we're treating the operating system as a container. That is 100% true.

That also means that operating system is an immutable image. So nobody can make any changes to your system. It is what it is. There's no changes. So once again, we've talked about the encryption aspect of this, but now we're really talking about what you know, you can't make changes. You can't cause harm. You're really locking it down to that point at, you know, by doing all of this. Um, you know, like I said, I don't know how important security is with you. I know you do a lot of, you know, you said entertainment, you know, over the in the airplanes, movies, and services and things like that.

00:20:20

Jason Anderson: I would imagine you have a lot of sensitive data, you know, credit cards or something like that that go through for charges. Um, you know, might be something to think about for your systems as you move forward.
Jason Dobyns: We have a cyber security team and they're they're they're rolling out tools all all over the place. Uh Crowd Crowd Strike things.
Jason Anderson: But it's
Jason Dobyns: Um we we had Rapid 7 before and they're I think we're getting rid of Rapid 7 this year and switching to some to another type of provider. But uh but yeah, so I mean we we if we get down to a topic like that, we can absolutely include some people from the cyber security side to for you to talk to them about that as
Jason Anderson: you know, if if that's something,
Jason Dobyns: well.
Jason Anderson: you know, if you want to give me some contact information or just make an introduction, I I would love to, you know, just have a conversation and see what see where they are and see if there's something we can help out with.

00:21:18

Jason Dobyns: Yep.
Jason Anderson: Um, there is definitely no harm in that. Um, but I kind of need to want to finish up through here. So, you said you had some questions. Um, but once again, highlighting image mode, you know, I can't tell you how much I love it, but it it provides the best of both worlds. Like I said, you still have the ability to do a yum update or a DNF update like you would in your traditional operating system, but you're still running in a container. So, nothing's really changing. If if I restart that container, it's going to go back to the same versions that I initially had it on. So you're combining the best of both of

these worlds and that's you know what we call image mode and you know this is a lot of repeat on the stuff that I've said along but it just provides you that simple path and you're Jason I talked to you I know you're an automation guy you're probably just like me I have to do it more than once I'm writing some sort of automation to do it because I don't want to have to remember how to do it again.

00:22:29

Jason Dobyns: That's that's exactly my like if almost all the time if I have to do something more than once I I automate it or I write a script to do it. Um I I used to write scripts that basically would get passed around on the servers and or the developer machines and that was so they could easily share code between each other and that was that was before git was like it is today.
Jason Anderson: Yep.
Jason Dobyns: That was back in like SVN days and some of some some stuff on subversion was a little more
Jason Anderson: Oh, I remember.
Jason Dobyns: a little trickier.
Jason Anderson: Yeah.
Jason Dobyns: So Yep.
Jason Anderson: My little checkouts and checkins and boom bring back
Jason Dobyns: So I I would sit and I'd watch what you know developers were doing sharing code and
Jason Anderson: nightmares.
Jason Dobyns: basically they would save the files on their system mess around in file explorer copy them to a network share then the other guy would go do the same thing and they do that a lot before doing
Jason Anderson: And they wouldn't work the

00:23:23

Jason Dobyns: their SVN commit right so so like at those points I then wrote
Jason Anderson: library.
Jason Dobyns: scripts and that would just pop the files there instead of them having to go manually find this the folder and the
Jason Anderson: Yeah.

Jason Dobyns: like So anyways, so yeah.
Jason Anderson: Absolutely.
Jason Dobyns: So now it's a little different,
Jason Anderson: Yeah.
Jason Dobyns: but
Jason Anderson: But yeah, I mean, but this is I mean, when you break it down to its most fundamental level, that's what what I'm talking about with this entire presentation with real 10, right? It's treating everything in that mindset. I don't want to have to figure out how to create snowflakes. I want to create a consistent environment that my users run on. I want to enforce policies that are going to keep my system secure and compliant. and I want to protect the developers from themselves because that's ultimately my job. So, you know, once again, bringing it all to all to the front of, you know, why I think RA 10 is awesome. Um, here's an example of an image mode file.

00:24:25

Jason Anderson: It's much like your traditional Docker build files or, you know, Podman build files. You call your base image wherever you want it to be. You install these very specific packages uh and dependencies that you want to run on that system. You add your application uh layer. You add your configuration files. Anything you want to run at startup and boom, your image will be built and done within a couple minutes.
Jason Dobyns: Is there a command that we can run to list what images or what containers are
Jason Anderson: So,
Jason Dobyns: run like this way? kind of like a Docker LS or something like that.
Jason Anderson: Uh there's so for it depends on how you're talking about this, right? So if you're using image mode, remember you're talking about the entire operating system.
Jason Dobyns: Yeah.
Jason Anderson: So yes, you can run containers on in image mode uh and have some sort of nested virtualization stuff going on there, but that's going to get kind of weird. Um what I would probably do in this particular scenario is maybe not use image mode but uh install traditional real 10 and use a utility like podman desktop or something like that to manage individual containers on a server.

00:25:51

Jason Dobyns: Gotcha.
Jason Anderson: Um and if you're going to go bigger than that obviously we need to start talking about Kubernetes platforms and you know scale container management at scale.
Jason Dobyns: Yeah. And kind of like like what I said earlier, the thing that I've been well that seems like the companies like it is how ephemeral the Kubernetes nodes are in the grand scheme of things. If you need to do an update on them, you can replace all of them and it moves the workloads with sometimes no downtime, sometimes a couple of seconds. But as soon as as soon as all the workloads are moved off of one host to a new host, it then can get rid of the old host. And that's that's an easy
Jason Anderson: You know,
Jason Dobyns: way.
Jason Anderson: you you know you can make it so there's absolutely zero downtime,
Jason Dobyns: Yeah.
Jason Anderson: right?
Jason Dobyns: But like I'm saying usually the the short down times are where it's one app and so then it
Jason Anderson: Okay.
Jason Dobyns: sees the pod. So I I I know now that there's different health checks you can add in the containers so that it doesn't think that the pod is moved until the new container is up and healthy.

00:26:57

Jason Dobyns: So that's definitely a a method, but I've I've seen some apps and it's mostly because even if the app is up,
Jason Anderson: Yeah.
Jason Dobyns: it's still doing other things in the background. So technically it thinks it's healthy in some cases, but it's still a few more seconds before it's all the way up and ready for,
Jason Anderson: Yeah.
Jason Dobyns: you know, communication or or requests being made to it,
Jason Anderson: Well,
Jason Dobyns: things like that. So I saw that a few times, but like I said, it was a couple of seconds at at the
Jason Anderson: yeah. I I just learned about that through,

Jason Dobyns: worst.
Jason Anderson: you know, certain applications that I saw re, you know, that said, hey, I don't care what the scenario is, I have I have to have at least one copy running.
Jason Dobyns: Yep.
Jason Anderson: So it would force a migration before it could delete that container.
Jason Dobyns: Yep.
Jason Anderson: Uh much like much like you're talking about
Jason Dobyns: And and the fix for those two is then uh not just running one container in that

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Jason Anderson: there.
Jason Dobyns: pod, having it have two, three, something like that, right?
Jason Anderson: So yep.
Jason Dobyns: And so then then it it does that juggle and if there is a container that doesn't respond to requests or or communication, it's not added to the ingress part until it's all the way ready. So I've seen I've seen those do well too.
Jason Anderson: Yeah, the
Jason Dobyns: But like I said, the only issue I ever had was the service was just running one.
Jason Anderson: only
Jason Dobyns: And I today these days I don't really consider one server or service running as production ready. You need sometimes multiple ones across multiple physical hosts.
Jason Anderson: Yeah.
Jason Dobyns: That way if there's any kind of anything that takes one host down, it shouldn't take the service down.
Jason Anderson: So your architecture that you just proposed actually could uh with spinning up your replicas. Um I'm just going to give you a scenario very quickly. If you when you are going through doing an update of Kubernetes, it first thing it is going to do is drain a node of resources, right?

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Jason Anderson: Everything that's running on there. If you're scheduling so many

replicas, it could even be two, right? It could potentially land on the same host. We don't know how that's going to land, right? That's all going to be up to the Kubernetes schedule.
Jason Dobyns: Yep.
Jason Anderson: So you start increasing those replica counts in these status of an upgrade, you can really break your upgrade to where it will fail because you can't migrate a pod because you've migrated everything else off and utilized your resources. So just something to be careful with because yes, you can actually break stuff.
Jason Dobyns: So, I think I think I I think it's been I I think it might have always been a feature, but now there's there's a way to do um you can add um node selector labels. And basically,
Jason Anderson: Yep.
Jason Dobyns: that's trying to tell it don't run this don't run these on the same
Jason Anderson: Run on this house.
Jason Dobyns: node. make sure they they that you're across nodes.
Jason Anderson: Yeah.
Jason Dobyns: And so again, like I said, in a in a small node environment with one or two nodes, that's much more possible, I think.

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Jason Anderson: Yeah.
Jason Dobyns: But when you're up to three or five or seven or, you know, me adding more nodes and then the update process is only going to try to do one node at a time, it kind of keeps everything still churning along, right? That's that's just my experience a little bit.
Jason Anderson: Absolutely. Um,
Jason Dobyns: So,
Jason Anderson: do you guys use Red Hat Insights?
Jason Dobyns: I don't think we use it. Um, it's been a while since I logged into my Red Hat account. Let me see. It didn't like my password. I'm going to have to reset it later. I was out from August till January.
Jason Anderson: Okay.
Jason Dobyns: Yeah, I I I had a very bad injury. I broke my skull.
Jason Anderson: What' you do?
Jason Dobyns: I I was Do you know what a onehe is?
Jason Anderson: No.
Jason Dobyns: It's a skateboard with a go-kart wheel in the middle and it bounces like a

Segway. So,
Jason Anderson: Okay.
Jason Dobyns: when you're at the top speed they're set to go to, they're set to go at I was going 22 miles an hour and you're asking it to go faster still, it gives up and noses down and it throws you off the front.

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Jason Dobyns: You then have to roll or run. I tried to roll or so I tried to run. I couldn't go from not running to running 22 miles an hour. I went to try to roll. My head hit the ground and popped in half. My skull popped in half. Yeah. So, I was in a medically induced coma for the next six weeks after that. Um, and then I got I got out of the hospital early November and then was going through a lot of physical occupational therapy stuff and came back to work in January. So, I'm I'm I'm I was very well taken care of. Um, but yeah,
Jason Anderson: It's I'm sorry.
Jason Dobyns: brain surgery is not not nice.
Jason Anderson: Yeah. Sorry to hear about the injury, but I it's awesome.
Jason Dobyns: Yeah.
Jason Anderson: You've made a recovery and everything is going well.
Jason Dobyns: Yep. Thank
Jason Anderson: Um well, so if you guys don't use Red Hat Insights,
Jason Dobyns: you.
Jason Anderson: um it has been changed to now something uh Lightseed.

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Jason Anderson: I forget exactly what it is. Red Hat is rebranding everything Lightede. Uh you'll see that coming out in future releases. Um, but essentially this allows you to administer all of your systems from the hybrid console. So as long as you register your system, it will report all the packages, all the system information, all of those things. So you have that single pane of glass to go through and see everything. You see your uh all your vulnerabilities, your updates are, you know, you can push updates through there. And yes, you can, as I said, even update image mode uh for certain things. Um, but it's everybody likes to reference the term single pane of glass. This is Red Hat giving you a single pane of glass to manage all of your systems. Um,

Jason Dobyns: Yep.
Jason Anderson: being that it's a free tool, I think that's kind of cool.
Jason Dobyns: I do know we do have I we probably do have this.
Jason Anderson: Um,
Jason Dobyns: Um I I have the only time I've been going into the Red Hat account last year was to look at how many licenses we were using and getting all getting everything registered and synced up.

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Jason Anderson: Okay. So, well, you know, like I said, I would highly recommend taking advantage of it.
Jason Dobyns: Sure.
Jason Anderson: It's a free tool. It's out there. You know, it can't hurt to spend an hour or two playing around with it. Um, you know, it's it's active work.
Jason Dobyns: Yeah, I'll definitely take a look through
Jason Anderson: Yeah. Um,
Jason Dobyns: it.
Jason Anderson: you know, once again, I I don't want to keep harping on these things, but we're tying it all together with the better decisions at build time with the hardened images. Now, you can give that complete security supply chain to your, you know, to your security officers or whatever. If you have that kind of level of security, it makes it so much easier to have all of those artifacts and you can, you know, re easily show it, document it, all of those great things. Um, from a security and compliance standard, it makes everybody's life easier because, as you know, they're going to say, "Oh my god, there's a RPM that you need to update and because it had this vulnerability and you're like, dude, it has a CDE score of two." um come on,

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Jason Anderson: you know. Um but this will help rectify those images and be able to roll it out faster um for those sorts of uh scenarios. Um
Jason Dobyns: I've been I've been doing this Crowd Strike image assessment deployment stuff and
Jason Anderson: and
Jason Dobyns: they it it's not listed as a CVE. It comes from a different um group that

does security stuff, but they don't like any Docker file that uses AD. It tell you to use copy. Every base OS image that I went through used AD because when you run add, it will extract a tar.gzip file and as part of the process, if you just use copy, you then have to run the commands in that container to extract that file and make sure the files are moved to where they need to go then. So, like I said, almost all the base images, base OS images that I I look through use ad and it came up with a complaint. and it showed it as medium and I it's because they think that um it's not good because AD can also be used to get a grab a URL like a file from a server and their determination was that file could be changed by uh an attacker or somebody not that you don't trust necessarily and that would affect the container.

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Jason Dobyns: So I I understand that side, but if your Docker file history just shows add and it adds a a zip file that's local to the project that the Docker build is running in, it's seems like that wouldn't hit that um that that security concern
Jason Anderson: Yeah, I I would tend to agree with you on that one.
Jason Dobyns: level
Jason Anderson: Um, so, uh, yeah, the image builder thing. Once again, I know you're a Packer shop. I'm not going to try and, uh, sell you, but this was exactly what I was talking about. They're going through saying install Realmd D, and it's going to say, "Hey, we see other people at you install the ADCLI." um when they install RealmD, you know, do you want to install that package? Once again, just a helpful thing when you're going through of, oh crap, I forgot a package. Let me go rebuild this image. While yes, it's easy to do, it's at that point becomes an annoyance.
Jason Dobyns: I've never seen AD CLI.
Jason Anderson: Um,
Jason Dobyns: We use um we use SSSD which I think does Keraros stuff and for then active directory login um to work but it's a little different.

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Jason Anderson: Yep.
Jason Dobyns: I I'll check out AD CLI. See if the join is easier with that.
Jason Anderson: Uh then we talked about the cryptography aspect of this. Um you know as I said it meets the FIPS cryptograph standards. We are the first ones to do it

which is pretty cool. Um and you know like I said it's not that hard to to implement and you know it's a great line down at the bottom. The best defense is a strong partner. Well, you got you guys got a strong partner here. That's our whole goal.
Jason Dobyns: Yeah.
Jason Anderson: Um, you know,
Jason Dobyns: Like
Jason Anderson: once again,
Jason Dobyns: that.
Jason Anderson: I'm rep reiterating myself, but I can't stress this enough. And I Jason, I think you really see the value here of we're really changing the mindset of how the traditional CIS admin thinks, right? It this is paradigm changing from the operating system perspective. Um, and it's been a long time since I got excited about a new Linux distro, and this one's got me excited.

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Jason Anderson: Whoops. Um, talked about the AI aspect of this. This, you know, once again, you can know you can host your models, you know, is safe, secure. You can import your data. So, you know that you're having good models. You can experiment with these on local systems. Um, you know, obviously you can throw GPUs in there. Um, you know, we talk about using Docker Desktop to manage your AI applications and containers on your systems to carve out specific resources for what you're trying to do. Um, you know, think of it one cool thing that uh I was actually thinking about, run an inference server and have all of your systems send their SIS lock. Imagine what you could do with that kind of information with AI as far as detecting anybody that did anything on your system. Ju I mean, just think about the possibilities for a second. I mean, you're you're taking it to a whole new level here. And you know, like I said, I don't get excited about new Linux OS's, but this this one's very different.

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Jason Anderson: Um, and that is the end of my spiel. Um on row
Jason Dobyns: I do like I'm absolutely going to try real 10. Um most likely I'm going to put it in my packer project at first just
Jason Anderson: 10.
Jason Dobyns: because most of the automation work is there. All I'll have to do is add,

you know, a RE10 ISO to to to the file system and then I'll try to just see if the same install method or that we use to install the things that we want on every golden image that we make with Packer.
Jason Anderson: Mhm.
Jason Dobyns: If it works the same from the Red Hat 9 method, uh, that'll be easy. If I have to change up some commands or package names, that'll be in that process where where I figure that out,
Jason Anderson: Yeah, but then you know I hate this this term as a CIS admin because you
Jason Dobyns: right?
Jason Anderson: already have to go through pain, but once you get to everything transition to image mode, think about how easier your updates and your upgrades will be.

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Jason Dobyns: Yeah. I mean I
Jason Anderson: I mean, it's just it's it's the light at the end of the tunnel kind of scenario.
Jason Dobyns: that's
Jason Anderson: But
Jason Dobyns: now what what do you think at this point like in the future with Redat 11 and 12 is that going to be a somewhat easy upgrade process hopefully. Okay.
Jason Anderson: Yes.
Jason Dobyns: I mean, we've got we've still got a lot of uh CentOS7 machines that we're paying for long-term extra support to get to get RPM updates. And that's the um part of the problem with the apps that run on those is they need an older version of Java and installing old Java on new Linux is not easy a lot. So, that's that's why they're still running because they're they're important machines. Um, and we're going to be working with that team sometime this year once when I get more of the AKS deployment completely finished and ready for production. Um, and we'll look at trying to containerize those applications and we'll see how that goes. But but that's why we're running some old old machines.

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Jason Dobyns: And there's there's a couple of other old flavors of of Linux uh running around 12. I I I don't know if we still have them. Do we still have any Red Hat 6 instances? Um, Sudil.
Sunil G: I don't think so.
Jason Dobyns: Okay. I know we got we we we got rid of so when we when we did the inventory project last year,
Sunil G: Even if we Yeah.
Jason Dobyns: we ended up decommissioning almost many of the old servers and just generating them a new
Sunil G: We have
Jason Dobyns: one and moving the application to the new ones, you know. So,
Sunil G: Yeah.
Jason Dobyns: I I know we went through that and I know that we do have a couple of ones that are dangling out there and it's going to take some some extra work to get those old apps. I think the company that made the apps that we use on those old ones that need the old Java, the company went out of business during the COVID lockdown or something like that and hasn't been

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Jason Anderson: You know
Jason Dobyns: available to make updates. So,
Jason Anderson: what?
Jason Dobyns: you know, but it's like you said, it's a it's an important system. It's it's one that collects data from all the airlines and it talks about the purchases PE not talks but it's data on anything purchased internet or media that was purchased and that kind of goes on in on our side and like contracts with those airlines on how to you know how to get paid the the percentage that we should. So that's why that's why we can't take those old machines down till we have something new. We tried to install that that we tried to install we tried to get them running on on Red Hat 9 or maybe eight and last year and um it was it didn't it didn't work. So that's that's going to be something we tackle once we start our new company year and couple couple weeks or month or two from now.
Jason Anderson: new company.
Jason Dobyns: No.
Jason Anderson: Did I hear
Jason Dobyns: So the the the corporate year here ends uh at is it at the end of is it the

end

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Jason Anderson: that?
Jason Dobyns: of March Sunil or it's the end of March is the end of the corporate year.
Sunil G: Yeah.
Jason Dobyns: So it's we we we're we're still on our all of our goals and plans for 2025 still corporate year 2025. So, we'll we'll be doing we'll be doing a bunch of planning and stuff in the next in the next couple of sprints and making plans and goals for the next corporate year that starts in April for us. So,
Jason Anderson: Well,
Jason Dobyns: yeah.
Jason Anderson: it I would love to be a part of those discussions. If there's anything we can do to help, um, please let me
Jason Dobyns: Um, like I said, I mean,
Jason Anderson: know.
Jason Dobyns: the first steps for me is to get a golden image of Red Hat 10 ready and we can we can launch a couple of those. Um, we've uh we've I I think most of the instances we've launched in the last year or so now are mostly Red Hat 9. Um, I know we have a few that are Oracle Linux9 and now it's because that team didn't want to chip in for the licenses.

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Jason Dobyns: So like we we we spread that cost out for those licenses to the because we have many development teams here different stuff. So, usually when they want a few more servers, if they only want one or two, usually I don't think they do that, but if they have like a dozen servers or something and they we would we would get there, but that team's budget to cover the licensing, you know. So, that's um I would call it a hurdle. It's just discussions that that have to go a couple different ways through through pass through different different
Jason Anderson: here.
Jason Dobyns: people. So,
Jason Anderson: All right. Uh well, in the link, I just posted the starting place for you with the real 10 UBI image.

Jason Dobyns: Awesome.
Jason Anderson: So, if you're going to build a golden path image, uh that should definitely be your starting place.
Jason Dobyns: Okay, I will absolutely check that out sometime very
Jason Anderson: All right.
Jason Dobyns: soon.
Jason Anderson: Um, did you guys want to talk about anything else or I mean I know Bill had a couple

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Jason Dobyns: Well, I
Jason Anderson: questions.
Bill Venella: Yeah, I'd love to uh but have to run to a different call.
Jason Anderson: Are we at time?
Bill Venella: Maybe we could set up a followup to understand some of your Kubernetes interest and then we can uh set up that topic for the the next call that we have.
Jason Dobyns: Once I'm done with the Crowd Strike um deployment to AKS, um I've I'm that's basically going to get it to the point where we can do our um readiness review for anything new we have to go through a production readiness review with cyber security team. And so once they sign off on that, I'm going to create a nonpro and a prod AKS in the the different Azure subscriptions for those and go forward. And then once we're done with that, then we could have a decent part of the conversation be showing you what where we're at today. And then hopefully you can tell us how open shift or show us how open shift could help us switch you know from just pure AKS to open shift in AKS or on prem.

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Jason Anderson: Okay.
Jason Dobyns: So that's I don't have any specific questions just yet,
Jason Anderson: Yeah.
Jason Dobyns: but like I said, I'm almost done with with the the the I call it the it's the shared um deployment. And so that's anything in the shared network is where you run services that something from prod and non-pro may need to communicate with, right? So it's going to run less stuff in shared than in prod or non-pro. So, but

Jason Anderson: Understandable. All right. Well, I need to jump too, but uh it's been great.
Jason Dobyns: yep.
Jason Anderson: Jason, I do have one ask and so I don't care who does it. I just need I hate coming into a meeting unprepared with no agenda. Um so if you guys could just let me know anything that you want to discuss. You know, think about it.
Jason Dobyns: Sure.
Jason Anderson: At minimum, I'm going to send you an email or bug you a week out saying, "Hey, what the hell do you want to talk about?" Pardon my language,

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Jason Dobyns: No,
Jason Anderson: but uh it's the
Jason Dobyns: it's okay. And uh obviously the the next one we'll see where I get
Jason Anderson: way.
Jason Dobyns: doing the the setup for our golden image of for for real 10 and then we'll if I have some progress on the AKS side.
Jason Anderson: Okay.
Jason Dobyns: Like I said, I think I think being able to show you where where we're at with our current AKS setup
Jason Anderson: Be more
Jason Dobyns: will and because I don't know exactly what specific questions a demo of
Jason Anderson: beneficial.
Jason Dobyns: Open Shift is awesome and I'm sure we'd like to see that, but like I don't know what questions I would ask without you guys having a better understanding of what I've done, what what we've set up in AKS.
Jason Anderson: Fair enough.
Jason Dobyns: So,
Jason Anderson: Okay.
Jason Dobyns: yeah.
Jason Anderson: All right. Well, let me know whenever you're ready. And you know, like I said, please any topics you guys want to discuss,
Jason Dobyns: Well,
Jason Anderson: let me know. I will be happy to bring in, you know, the appropriate teams. You know, whether it be a specific topic on RE or anything else,
Jason Dobyns: sure.
Jason Anderson: I don't really care. Just I'm here to make your guys' life easier and I

want you to find value out of these meetings. Otherwise, you'll stop coming.
Jason Dobyns: Well,
Jason Anderson: All right.
Jason Dobyns: thank you very much, Jason.
Jason Anderson: Yep. You guys have
Jason Dobyns: I liked your I liked your uh I liked what do you what do you call I like that I
Jason Anderson: good
Jason Dobyns: like that whole demo not demo I like that whole talk you just had with us about the image it's uh again it does seem like the a better path or an easier path forward and if the upgrade from like real 10 to 11 is easy or if you run it in image mode that is something that will be easy to discuss with the people here that would make those types of decisions

Transcription ended after 00:49:57